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FIRST-ORDER CONTROL USING OPTIMAL TRANSPORT SUPERVISION FOR IMITATION FROM OBSERVATIONS

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Summary

This paper proposes FOCUS, a first-order imitation-from-observations (ILfO) algorithm that leverages differentiable physics to obtain low-variance, pathwise policy gradients from state-only demonstrations without temporal alignment. The method aligns short agent rollouts to multiple sampled expert segments via entropy-regularized optimal transport (OT) and aggregates them with a differentiable soft best-of-K objective to emphasize locally compatible phases, while a learned critic bootstraps value beyond the local horizon. Across four contact-rich DFlex locomotion tasks, including a 152-actuator musculoskeletal humanoid, FOCUS improves sample efficiency and wall-clock stability over adversarial ILfO and differentiable imitation baselines.

Strengths

Technical novelty and innovation

- Combines three ideas in a well-integrated way for ILfO: (i) local OT over short rollouts to relax temporal alignment; (ii) a soft best-of-K relaxation to select locally compatible expert phases while retaining differentiability; and (iii) short-horizon first-order actor–critic with value bootstrapping to propagate credit beyond the local window.
- Differentiating through Sinkhorn-regularized OT to construct per-step shaped rewards and backpropagating through the differentiable simulator is a principled way to reduce variance compared to adversarial or purely black-box OT rewards.
- Reward-shaping transformations of local OT costs to stabilize critic targets are a practical and well-motivated design choice.

Experimental rigor and validation

- Evaluations on four DFlex tasks, including the challenging SNU Humanoid with 152 muscle actuators, demonstrate strong performance and stability in contact-rich settings where first-order methods are most compelling.
- Comparisons include diverse baselines: adversarial ILfO (GAIfO, MAAD, SAM/DAC), Wasserstein/OT-based rewards (PWIL, OOPS, LWAIL), and a differentiable-physics IL baseline (ILD), plus ablations (FOCUS-L2, OT variants, critic/no-critic, zero-order variants), and diagnostics (runtime and K sensitivity in appendices).
- Reporting uses expert-normalized return with IQM and CIs; both interaction and wall-clock comparisons are shown.

Clarity of presentation

- The method is clearly described with a compact algorithm, equations detailing the OT objectives, best-of-K relaxation, and the actor–critic surrogate; design choices (segment length, horizon, temperature, shaping) are explicit.
- Limitations are candidly discussed (need for differentiable dynamics, local rather than global alignment).

Significance of contributions

- Addresses an important gap: ILfO in contact-rich domains where adversarial rewards and black-box OT matching often struggle with credit assignment and stability.
- Demonstrates that first-order gradients, paired with local OT

alignment, can scale to very high-dimensional musculoskeletal control from state-only demonstrations without teacher forcing or pretraining.

Weaknesses

Technical limitations or concerns

- The best-of-K objective is inherently local and can be mode-seeking: it may under-cover rare phases or encourage fragmentary matches in non-periodic tasks; no mechanism enforces global ordering or coverage beyond the local window.
- The critic learns under a non-Markovian shaped reward (weights and transport plans depend on the entire window), so it is a surrogate rather than a consistent value function for a fixed per-step reward; potential bias is acknowledged but not analyzed theoretically.
- Gradients through a finite number of Sinkhorn iterations can introduce approximation bias; the paper lacks a study of sensitivity to ε and iteration count on stability and learning outcomes.

Experimental gaps or methodological issues

- While zero-order variants are compared (SAC/PPO), there is no “same-backbone, no-simulator-gradients” control (e.g., SAPO using the same shaped reward with stop-gradient through dynamics) to isolate the benefit of simulator derivatives.
- No direct comparison to temporally constrained sequence objectives (e.g., TemporalOT, ORCA) that explicitly enforce ordering/coverage; this is particularly relevant to non-periodic or multi-stage tasks.
- Generalization beyond DFlex periodic locomotion is not evaluated (e.g., manipulation, non-periodic tasks, sparse subgoals), limiting claims of broader applicability.

Clarity or presentation issues

- Some choices (segment length L, horizon H, K) are justified empirically but the trade-offs are not discussed in depth; a concise

guidance table or sensitivity study in the main text would aid practitioners.

- The relationship between optimizing the softmin-aggregated per-step costs (with gradients through weights) and the log-sum-exp surrogate is clarified in the appendix but could be summarized more prominently.

Missing related work or comparisons

- Temporal alignment/coverage methods (TemporalOT, ORCA) are mentioned, but comparisons are not presented; given the paper's emphasis on misalignment, such baselines would strengthen the case and contextualize the "local selection vs. global ordering" trade-off.
- Additional differentiable imitation baselines (e.g., GDTW-IL) are discussed but not empirically compared; even a partial comparison on smaller tasks could be informative.

Detailed Comments

Technical soundness evaluation

- The use of entropy-regularized OT with Sinkhorn is standard and well-suited to differentiable training. Differentiating through the transport plan and through the softmin weights is sound, and the paper correctly notes that the actor update does not equal the gradient of the log-sum-exp minimum over full trajectories—this is a shaped local surrogate optimized with bootstrapping.
- The critic approximation under non-Markovian short-horizon rewards is a practical compromise; similar mixed-order methods (SHAC, SAPO) motivate it. A small analysis of bias/variance introduced by the window-level dependence would be useful, but the empirical stability suggests the surrogate is effective.
- Reward-shaping transforms are monotone and preserve local ranking, but they may alter global policy ordering. The authors position them as shaping for critic stability rather than as a principled potential-based shaping; this is acceptable but should be emphasized when interpreting improvements.

- Finite Sinkhorn iterations and fixed ε can bias gradients; stability and wall-clock diagnostics are provided, yet sensitivity plots to ε/ iterations would help practitioners pick settings and understand robustness.

Experimental evaluation assessment

- The task suite is appropriate for demonstrating the value of first-order gradients in contact-rich control, and the SNU Humanoid results are compelling.
- Baseline breadth is strong. However, to more cleanly isolate first-order benefits, a control using the same SAPO backbone without differentiating through dynamics (e.g., treat r_t^FOCUS as an extrinsic reward and stop-grad through dynamics) would close a methodology gap; the current ZO comparisons also change the RL algorithm.
- The best-of-K mechanism is central. While the paper explores K in appendices and shows diagnostic alignment improvements, a main-text ablation of K=1 vs K>1 and “no gradient through weights” vs “with gradient through weights” would better quantify the contribution of soft best-of-K selection.
- Reporting is careful: IQM with CIs, step and wall-clock axes, consistent architectures via MINERAL where possible. Runtime breakdowns (batched Sinkhorn cost as a fraction of total) are appreciated; still, summarizing absolute per-iteration time and scaling with K and L in the main paper would strengthen the practical message.

Comparison with related work (using the summaries provided)

- Relative to adversarial/occupancy methods (MAAD, PWIL, LWAIL, OOPS, DILO/SMODICE variants), FOCUS avoids discriminator instability and high-variance updates by exploiting simulator derivatives and local OT; empirical results corroborate better stability and efficiency in DFlex tasks.
- Compared to differentiable imitation approaches (ILD, DiffMimic, GDTW-IL), FOCUS removes the need for teacher forcing or aligned initialization and addresses temporal misalignment with soft best-of-K local OT, while using short horizons to mitigate contact-gradient brittleness—an advance over full-trajectory alignment in contact-rich regimes.
- Temporal/coverage-focused methods (TemporalOT, ORCA)

address global ordering—an orthogonal dimension where FOCUS is currently local. Incorporating ordered coverage could broaden applicability to non-periodic or multi-stage tasks.

- Recent first-order control advances (SHAC, AHAC, SAPO, ABPT, DMO) highlight the benefits and pitfalls of gradients in contact. FOCUS builds on SHAC/SAPO's stability mechanisms; exploring AHAC-style adaptive horizons or ABPT-style gradient mixing may further reduce brittleness when OT signals conflict with contact-induced gradient spikes.

Discussion of broader impact and significance

- The approach meaningfully improves ILfO in settings where state-only demos are available and differentiable physics is accessible, advancing the frontier on high-DOF contact-rich control.
- Practical impact could be substantial in simulation-intensive robotics pipelines. Real-world transfer will require bridging the differentiability assumption, e.g., via learned world models or decoupled gradients as in DMO; the paper acknowledges this.
- Risks are limited: the method does not require access to actions or human-annotated rewards; however, care is needed for safety when transferring policies trained with shaped non-Markovian rewards to real systems.

Questions for Authors

1. How sensitive is performance to the Sinkhorn parameters (ε, iteration count)? Could you provide a brief sensitivity analysis and guidance on choosing these hyperparameters across tasks?
2. Can you isolate the effect of simulator derivatives by running a control that uses the same SAPO backbone but stops gradients through the dynamics (i.e., treats r_t^FOCUS as an extrinsic reward optimized purely with model-free gradients)?
3. What is the contribution of differentiating through the softmin weights w_k? Do results change if you stop gradients through w_k (i.e., treat weights as fixed at each update), or if you use uniform averaging over the K segments?

4. How do K, H, and L trade off in practice? Are there diminishing returns beyond certain K, and how does runtime scale with K·L·H? A concise main-text table with absolute timings would help practitioners.
5. Have you tried adding simple temporal constraints (e.g., a diagonal mask as in TemporalOT) or coverage terms (ORCA-like) to reduce fragmentary matches on non-periodic tasks? If so, how did this interact with best-of-K?
6. When demonstrations are few or contain rare phases, best-of-K could repeatedly select common phases. Would adding a coverage prior or a curriculum over start indices u help? Any observations from your single-trajectory ablation?
7. You focus on state–state transitions. Did you try learned representations (e.g., LWAIL-style dynamics-aware encoders) as the ground metric for OT, and if so, how did that affect stability and wall-clock?

Overall Assessment

Estimated Score: **6.5/10** (Calibrated to ICLR scale)

This paper presents a compelling and practically useful contribution to ILfO in differentiable simulators. The core idea—turning temporally unaligned state-only demonstrations into short-horizon differentiable supervision through local OT and soft best-of-K selection—addresses two key pain points: temporal misalignment and unstable long-horizon gradients in contact-rich dynamics. The empirical results on DFlex, especially the SNU Humanoid, convincingly demonstrate improved sample efficiency and stable training over strong adversarial and OT-based baselines, and over a differentiable imitation baseline (ILD). While the approach is a thoughtful synthesis of existing building blocks (OT with Sinkhorn, short-horizon first-order actor–critic), the integration is novel in the ILfO context and yields clear benefits. Limitations are mostly about scope (differentiable simulators, periodic locomotion, local alignment) and can be credibly addressed in future work. I recommend acceptance. The paper would be strengthened further by an isolated ablation of simulator-derivative benefits with the same backbone, a short Sinkhorn sensitivity study, and an initial exploration of

ordering/coverage constraints for non-periodic tasks.

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